MATH120-24S2 Discrete Mathematics

Assignment 8

Due: 11:59pm Friday 27 September
Submit a single PDF document (1 file only) to the submission link in Learn. It is your responsibility to ensure that your submission is clear and easily readable, and that you upload the correct file by the due date.

1. Determine whether or not the following functions are one-to-one, onto, both or neither. Justify your answer with a proof or counterexample.

   (a) $f : \mathbb{Z} \to \mathbb{Z}$ given by $f(n) = 2n + 3$.

   (b) $g : \mathbb{R} \to \mathbb{R}$ given by $g(x) = x^3$.

   (c) $h : \mathbb{Z} \to \mathbb{Z}$ given by $h(n) = \begin{cases} \frac{n}{2}, & n \text{ even} \\ \frac{n-1}{2}, & n \text{ odd}. \end{cases}$

   Solution:

   (a) f is one-to-one and not onto. To see that f is one-to-one, suppose $f(n) = f(m)$, then $2n + 3 = 2m + 3$ and hence $2n = 2m$ so $n = m$.
   To see that f is not onto, consider 2 in the codomain $\mathbb{Z}$. Since n is an integer, there is no solution to $2n + 3 = 2$.

   (b) g is onto and one-to-one. It is a bijection.

   To see that g is one-to-one, suppose $g(x) = g(y)$, then $x^3 = y^3$ and hence $x = y$. *Note that this argument holds for all odd powers, but not even powers. Why?*

   To see that g is onto, we note that for all $y \in \mathbb{R}$, $\sqrt[3]{y} \in \mathbb{R}$ and hence $g(\sqrt[3]{y}) = y$.

   (c) h is onto but not one-to-one.

   One-to-one: note that $h(3) = 1$ and $h(2) = 1$ but $3 \neq 2$. There are many other examples, but this demonstrates that h is not one-to-one.

   Onto: consider $m \in \mathbb{Z}$. We have that $2m$ is even, so $h(2m) = 2m/2 = m$, and hence, h is onto.

2. Prove the following cardinality facts by constructing an appropriate bijection.

   (a) $|4\mathbb{Z}| = |3\mathbb{Z}|$.

   (b) $\mathbb{N} \times \{0, 1, 2\}$ is countable.

   Solution:

   (a) Let $f : 4\mathbb{Z} \to 3\mathbb{Z}$ be given by $f(n) = \frac{3n}{4}$. to see that this is a function we note that if $n \in 4\mathbb{Z}$ then there is some $k \in \mathbb{Z}$ so that $n = 4k$ and hence $f(n) = 3k \in 3\mathbb{Z}$.
   f is one-to-one: Suppose $f(n) = f(m)$, then $\frac{3n}{4} = \frac{3m}{4}$ and hence $n = m$.
   f is onto: let $b \in 3\mathbb{Z}$, then there is some $\ell \in \mathbb{Z}$ so that $b = 3\ell$. Then $a = 4\ell \in 4\mathbb{Z}$ and it follows that $f(a) = \frac{3 \cdot 4\ell}{4} = 3\ell = b$.
   Therefore f is a bijection and hence $|4\mathbb{Z}| = |3\mathbb{Z}|$.

   (b) $\mathbb{N} \times \{0, 1, 2\}$ is countable. There are many possible ways to show that $\mathbb{N} \times \{0, 1, 2\}$ is countable. One of them is the following. Consider the map $f : \mathbb{N} \times \{0, 1, 2\} \mapsto \mathbb{N}$ given by $f(n, i) = 3n + i$. We first show that this function is onto. Let $z \in \mathbb{N}$, and let $j \in \{0, 1, 2\}$ be equal to the value of z mod 3, then $z - j$ is a multiple of 3, say $z - j = 3k$, and hence, $f(k, j) = 3k + j = z$. This shows that f is onto.

   Now suppose that $f(n, i) = f(n', i')$, then $3n + i = 3n' + i'$. Considering this equation modulo 3, we obtain that $i = i'$ modulo 3. Now i and i' are both elements in $\{0, 1, 2\}$, so they can only equal modulo 3 if they are equal. We obtain that $i = i'$, and from $3n + i = 3n' + i$, it follows that $n = n'$ as well. So $f(n, i) = f(n', i')$ implies $(n, i) = (n', i')$ which shows that f is one-to-one.

   We have now given a bijection between the sets $\mathbb{N} \times \{0, 1, 2\}$ and $\mathbb{N}$, so we have shown that $\mathbb{N} \times \{0, 1, 2\}$ and $\mathbb{N}$ have equal cardinality. We know that $\mathbb{N}$ is countable because it is a subset of $\mathbb{Z}$ which is countable. This shows that $\mathbb{N} \times \{0, 1, 2\}$ is countable as well.

3. Let A and B be finite sets with $|A| = |B|$, and let $f : A \to B$ be a function. Prove that if f is one-to-one then f is onto.

   Solution:

Assume f is one-to-one and let $|A| = |B| = n$. The let $A = \{a_1, a_2, \ldots, a_n\}$ and $B = \{b_1, b_2, \ldots, b_n\}$. Since f is one-to-one, we note that $f(a_i) \neq f(a_j)$ for all $i, j \in \{1, \ldots, n\}$ such that $i \neq j$. Therefore $f(a_1), f(a_2), \ldots, f(a_n)$ are n distinct elements in B. However there are n elements in B, namely $b_1, b_2, \ldots, b_n$. It follows that for each $b_\ell \in B$, there is some element $a_k \in A$ so that $f(a_k) = b_\ell$.